Semi-Supervised DeepSDF with Eikonal Regularization and Fourier Positional Encoding

Yu Taek Lee
College of Computing and Data Science
Nanyang Technological University
Singapore
G2505374L
yutaek001@e.ntu.edu.sg

***Abstract*—Neural implicit representations such as DeepSDF learn continuous signed distance functions for 3D shape reconstruction, but require dense signed distance supervision for every training point. We investigate whether Eikonal regularization, which enforces the unit gradient norm constraint on unsupervised points, can reduce this supervision requirement. We train DeepSDF on 300 ShapeNet shapes at supervision ratios from 5% to 100% and evaluate with Chamfer Distance and Normal Consistency. Our experiments suggest that 10% supervision can remain competitive with or outperform the single-seed fully-supervised baseline in this setting: the Eikonal-regularized model achieves a 3-seed mean Chamfer Distance of 0.0496 compared to 0.0593 for the baseline. We additionally test Fourier positional encoding at two frequency levels, finding that it consistently degrades reconstruction quality across all supervision levels, collapsing Chamfer Distance to approximately 0.14 regardless of label availability or additional regularization. These results suggest that geometric priors are more effective than spectral input transformations for label-efficient 3D reconstruction.**

***Index Terms*—DeepSDF, signed distance functions, Eikonal regularization, semi-supervised learning, positional encoding, 3D reconstruction**

I. Introduction

Learning 3D shape representations from data is a central problem in computer vision with applications in robotics, autonomous driving, and virtual reality. Traditional approaches such as voxel grids and point clouds suffer from resolution limitations and lack of surface continuity. Neural implicit representations address these issues by parameterizing shapes as continuous functions—typically signed distance functions (SDFs)—that can be queried at arbitrary resolution [1].

DeepSDF [1] learns a neural network $f_\theta(z, x) \rightarrow s$ that maps a latent code z and a 3D coordinate x to a signed distance value s. Through an auto-decoder framework, per-shape latent codes are jointly optimized with the network weights, enabling the model to represent a distribution of shapes. However, DeepSDF requires dense signed distance supervision: every training point must have a ground-truth SDF value, which is expensive to obtain for real-world 3D data.

This supervision bottleneck motivates semi-supervised approaches that leverage geometric priors to reduce label requirements. The Eikonal equation $\|\nabla_x f(x)\| = 1$ is a fundamental property of signed distance functions [2]. Enforcing this constraint on unsupervised points—those without SDF labels—provides a training signal derived purely from the geometry of the problem. If effective, this could substantially reduce the annotation effort for training neural implicit representations.

Separately, Fourier positional encoding (PE) has been shown to help multi-layer perceptrons (MLPs) learn high-frequency functions by mapping low-dimensional inputs to higher-dimensional sinusoidal features [3], [4]. While this technique is standard in Neural Radiance Fields (NeRF), its interaction with SDF learning—particularly in semi-supervised settings—is less understood.

In this work, we systematically investigate two questions:

1) Can Eikonal regularization reduce the need for signed distance supervision in DeepSDF, and by how much?
2) Does Fourier positional encoding help or hurt reconstruction quality in this semi-supervised SDF setting?

We conduct 12 experiments on 300 ShapeNet shapes across supervision ratios from 5% to 100%, with and without Eikonal regularization, Fourier PE at two frequency levels, and second-order smoothness regularization. Our key findings are:

- Eikonal regularization allows **10% supervision to remain competitive**: the regularized model (3-seed mean) matches or outperforms the single-seed fully-supervised baseline in Chamfer Distance.
- Fourier positional encoding is **consistently harmful**: all PE configurations collapse reconstruction to approximately 2.4–2.8× worse than baseline, regardless of supervision level or additional regularization.

II. Related Work

A. Neural Implicit Representations

Neural implicit representations encode 3D geometry as the level set of a learned function. Occupancy Networks [8] predict binary inside/outside occupancy, while DeepSDF [1] regresses continuous signed distance values, enabling gradient-based surface extraction via marching cubes [7]. DeepSDF's auto-decoder framework avoids the need for an encoder by directly optimizing per-shape latent codes during training. While effective, it requires dense SDF supervision at every training point.

B. Geometric Regularization

The Eikonal equation $\|\nabla_x f(x)\| = 1$ is a necessary condition for valid signed distance functions. Gropp et al. [2] introduced Implicit Geometric Regularization (IGR), demonstrating that enforcing this constraint during training improves surface quality and enables learning from raw point clouds without SDF labels. This approach has been adopted in subsequent work on neural implicit surfaces [9], [10]. In our work, we apply Eikonal regularization specifically to reduce the number of supervised training points, quantifying the label efficiency gains across multiple supervision ratios.

SIREN [5] takes a different approach to geometric regularization by using periodic activation functions (sinusoidal) instead of ReLU, which naturally satisfies certain differential equation constraints. However, SIREN requires architectural changes, whereas Eikonal regularization can be applied as a loss term to standard ReLU networks.

C. Positional Encoding

Fourier positional encoding was introduced in NeRF [3] to overcome the spectral bias of MLPs—their tendency to learn low-frequency functions first [11]. By mapping input coordinates through sinusoidal functions at multiple frequency bands, $\gamma(x) = [\sin(2^0\pi x), \cos(2^0\pi x), \ldots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x)]$, the network can represent high-frequency variations. Tancik et al. [4] provided a theoretical framework for this approach based on the Neural Tangent Kernel.

While Fourier features are standard in radiance field methods, their application to SDF learning is less established. NeRF operates in a supervised regime where every ray has a target color, and the network is evaluated primarily at points along training rays. In contrast, SDF evaluation via marching cubes requires querying the network on a dense grid that includes regions far from the training data distribution, creating an extrapolation challenge for high-frequency encodings.

III. Method

A. Problem Formulation

Given a dataset of 3D shapes, we aim to learn a function $f_\theta : \mathbb{R}^{d_z} \times \mathbb{R}^3 \to \mathbb{R}$ that maps a shape-specific latent code $z_i \in \mathbb{R}^{d_z}$ and a spatial coordinate $x \in \mathbb{R}^3$ to the signed distance to the shape's surface. For each shape i, we have:

- **Supervised points**: $\{(x_j, s_j)\}_{j=1}^{N_s}$ with ground-truth SDF values
- **Unsupervised points**: $\{x_k\}_{k=1}^{N_u}$ with coordinates only

During preprocessing, each shape is sampled to produce a fixed pool of 250,000 supervised points (with SDF labels) and a separate pool of 250,000 unsupervised points (coordinates only). The supervision ratio r controls the fraction of the supervised pool that is retained: at $r = 0.1$, only 25,000 supervised points are used, while all 250,000 unsupervised points remain available. At $r = 1.0$ the full supervised pool is used alongside the full unsupervised pool. Each training batch samples half its points from the supervised set and half from the unsupervised set. We investigate $r \in \{0.05, 0.10, 0.50, 1.0\}$.

B. Architecture

We use an 8-layer MLP with hidden dimension 512, following the DeepSDF architecture [1]. A skip connection concatenates the input to the activations at layer 4. Each shape is associated with a learnable latent code $z_i \in \mathbb{R}^{256}$, initialized from $\mathcal{N}(0, 0.01^2)$ and jointly optimized with the network weights (auto-decoder). The final layer produces a scalar SDF value without activation, with weights initialized from $\mathcal{N}(0, 1/\sqrt{512})$ and bias set to 0.1 to approximate a sphere.

When Fourier PE is enabled, input coordinates are transformed as:

$$\gamma(x) = \left[\sin(2^0\pi x), \cos(2^0\pi x), \ldots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x)\right] \tag{1}$$

where L is the number of frequency levels. This maps each 3D coordinate from $\mathbb{R}^3$ to $\mathbb{R}^{6L}$. The network input becomes $[z_i; \gamma(x)]$ instead of $[z_i; x]$. We test $L = 4$ (24-dimensional) and $L = 6$ (36-dimensional).

C. Loss Functions

The total training loss combines four terms:

$$\mathcal{L} = \mathcal{L}_{\text{sdf}} + \lambda_{\text{eik}}(t) \cdot \mathcal{L}_{\text{eik}} + \lambda_z \cdot \mathcal{L}_z + \lambda_{\text{2nd}} \cdot \mathcal{L}_{\text{2nd}} \tag{2}$$

SDF loss ($\mathcal{L}_{\text{sdf}}$): L1 regression on supervised points only:

$$\mathcal{L}_{\text{sdf}} = \frac{1}{N_s} \sum_{j=1}^{N_s} |f_\theta(z_i, x_j) - s_j| \tag{3}$$

Eikonal loss ($\mathcal{L}_{\text{eik}}$): Enforces $\|\nabla_x f\| = 1$ on all points (supervised and unsupervised), providing geometric regularization from unlabeled data:

$$\mathcal{L}_{\text{eik}} = \frac{1}{N_s + N_u} \sum_k (\|\nabla_x f_\theta(z_i, x_k)\| - 1)^2 \tag{4}$$

Latent regularization ($\mathcal{L}_z$): Prevents latent code drift:

$$\mathcal{L}_z = \|z_i\|^2 \tag{5}$$

Second-order loss ($\mathcal{L}_{\text{2nd}}$): Optionally penalizes the divergence of the gradient field to encourage smoother surfaces:

$$\mathcal{L}_{\text{2nd}} = \frac{1}{N} \sum_k |\nabla \cdot \nabla_x f_\theta(z_i, x_k)| \tag{6}$$

D. Eikonal Warmup Schedule

The Eikonal weight uses a linear warmup to prevent the gradient norm constraint from dominating before the network learns basic geometry from $\mathcal{L}_{\text{sdf}}$:

$$\lambda_{\text{eik}}(t) = \lambda_{\text{eik}} \cdot \min\left(1, \frac{t}{T_w}\right) \tag{7}$$

where T_w is ratio-dependent: 100 epochs for $r \geq 0.5$, 150 epochs for $r = 0.1$, and 200 epochs for $r = 0.05$. Lower supervision ratios require longer warmup because fewer supervised points provide weaker initial geometry signal.

E. Training Procedure

Training iterates over all shapes per epoch, sampling a batch of 16,384 points per shape (split evenly between supervised and unsupervised). Gradients are accumulated across all shapes and applied once per epoch. We use the Adam optimizer with learning rate 5×10^{-4}, stepped by 0.5× every 500 epochs, and gradient clipping at $\max_norm = 1.0$.

Default hyperparameters are $\lambda_{\text{eik}} = 0.1$, $\lambda_z = 10^{-4}$, and $\lambda_{\text{2nd}} = 0$ (disabled unless specified). All models train for 3,000 epochs.

IV. Experiments

A. Dataset

We use 300 shapes from ShapeNet [6] spanning three categories: airplane, chair, and table. Raw meshes are first converted to watertight manifolds, then sampled to produce 250,000 supervised points (with ground-truth SDF values) and 250,000 unsupervised points (coordinates only) per shape. During training, shapes are split 75%/25% (225 train, 75 validation) for loss monitoring. Evaluation attempts to reconstruct all 300 shapes (both training and validation); however, mesh extraction occasionally fails for individual shapes (up to 41 failures in some runs), and reported metrics are computed only over successfully reconstructed shapes. Because evaluation includes training shapes, the reported metrics reflect fitting quality and should not be interpreted as generalization to unseen shapes.

For reduced supervision experiments, supervised points are subsampled to the target ratio (e.g., 10% retains 25,000 supervised points) while all 250,000 unsupervised points remain available for Eikonal regularization.

B. Evaluation Metrics

We evaluate reconstructed meshes extracted via marching cubes [7] at resolution 128^3 (overriding the codebase default of 256^3 to reduce evaluation time):

Chamfer Distance (CD): Bidirectional mean squared nearest-neighbor distance between 30,000 points sampled from predicted and ground-truth meshes. Lower is better.

Normal Consistency (NC): Mean absolute dot product between normals of matched point pairs. Higher is better (maximum 1.0).

Volumetric IoU was omitted due to computational constraints.

C. Experiment Design

Table I summarizes our 12 experiments. We systematically vary supervision ratio (r), Eikonal regularization, and Fourier PE to isolate each factor's contribution. Key comparisons include EXP-04 vs. EXP-01 (label efficiency with Eikonal) and EXP-06 vs. EXP-04 (PE effect at low supervision). Multi-seed runs (3 seeds) are conducted for EXP-04 and EXP-06 to assess variance.

All experiments are trained on NVIDIA A40 GPUs for 3,000 epochs (~4–6 hours per run). We use identical hyperparameters across experiments except for the specified variations.

TABLE I
Experiment design. All experiments use Eikonal regularization except EXP-01 (vanilla baseline).

ID	r	Eik	PE	Extra	Purpose
01	100%	–	–	–	Baseline
02	100%	✓	–	–	Eikonal effect
03	50%	✓	–	–	Half labels
04	10%	✓	–	–	Low-label (3 seeds)
05	5%	✓	–	–	Minimal labels
06	10%	✓	L=6	–	PE effect (3 seeds)
07	5%	✓	L=6	–	PE at min. labels
08	10%	✓	L=6	$\mathcal{L}_{\text{2nd}}$	Can L_{2nd} save PE?
09	100%	✓	L=6	–	PE quality ceiling
10	100%	✓	L=4	–	Lower PE freq
11	10%	✓	L=4	–	Lower PE, low-label
12	5%	✓	L=4	–	Lower PE, min. label

TABLE II
Main results. Single-seed rows report the per-shape mean over successfully evaluated shapes (259–300 depending on run; see Section V-F). Multi-seed rows (EXP-04, EXP-06) report the mean ± std of per-seed means *across 3 seeds*; these are not directly comparable to single-seed entries. Evaluation includes both training and validation shapes (Section IV-A).

ID	Config	CD (↓)	NC (↑)
Without Positional Encoding			
01	100%, no Eik	0.0593	0.5522
02	100% + Eik	0.0543	0.5920
03	50% + Eik	0.0534	0.5924
04	10% + Eik (3-seed)	**0.0496**±0.010	**0.5987**±0.020
05	5% + Eik	0.0509	0.5766
With Fourier PE (L = 6)			
06	10% + Eik + PE (3-seed)	0.1399±0.003	0.5078±0.002
07	5% + Eik + PE	0.1448	0.5070
08	10% + Eik + PE + L_{2nd}	0.1443	0.5045
09	100% + Eik + PE	0.1450	0.5027
With Fourier PE (L = 4)			
10	100% + Eik + PE	0.1401	0.5078
11	10% + Eik + PE	0.1427	0.5067
12	5% + Eik + PE	0.1399	0.5078

V. Results

A. Eikonal Regularization Enables Label Reduction

Table II presents the main quantitative results. The vanilla DeepSDF baseline (EXP-01) achieves CD = 0.0593 with 100% supervision and no Eikonal regularization. Adding Eikonal at full supervision (EXP-02) immediately improves CD by 8.4% to 0.0543 and NC by 7.2% to 0.5920.

The most striking result is that *reducing* supervision to 10% with Eikonal (EXP-04) yields a 3-seed mean CD of 0.0496, competitive with and—in two of three seeds—better than the single-seed baseline EXP-01 (CD = 0.0593). Seeds 123 and 456 individually outperform EXP-01, while seed 42 is slightly worse (0.0609). Because this compares a seed-averaged value against a single run, the 16.4% gap should be interpreted cautiously. Even at 5% supervision (EXP-05, single seed), CD = 0.0509 remains 14.2% better than EXP-01.

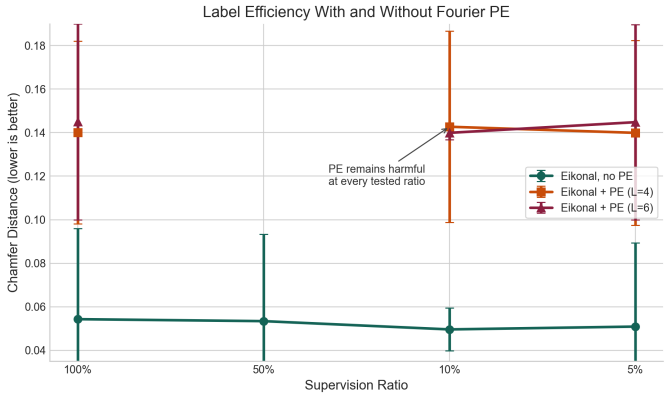

Fig. 1. Label efficiency: Chamfer Distance vs. supervision ratio. Low-supervision Eikonal models (green) remain competitive with or outperform full supervision, while PE models (orange, red) remain in a high-error regime regardless of supervision level.

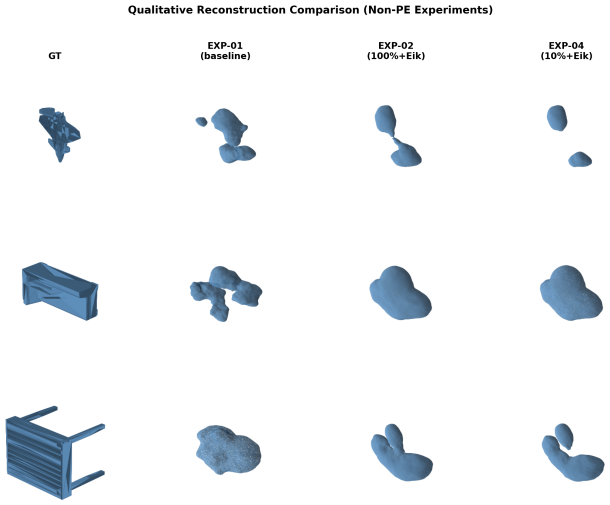

Fig. 2. Qualitative reconstruction comparison for non-PE experiments. Rows: airplane, chair, table. Columns: ground truth, baseline (EXP-01), Eikonal at 100% (EXP-02), Eikonal at 10% (EXP-04). All reconstructions are extracted via marching cubes at 128^3 resolution. PE experiment meshes are omitted (see Section V-F).

Fig. 1 illustrates this label efficiency curve. The non-monotonic pattern—where low-supervision Eikonal models remain competitive, with the best results appearing at 5–10% supervision in our runs—suggests that dense SDF supervision may act as a weaker regularizer than the Eikonal constraint. One plausible explanation is that the network overfits to noise in the SDF values when labels are abundant, while Eikonal regularization provides a geometric prior that more than compensates for the loss of direct supervision. However, confirming this hypothesis would require further analysis (e.g., per-shape error distributions or training-curve ablations).

B. Fourier PE Degrades Reconstruction

All seven PE experiments (EXP-06 through EXP-12) produce CD in the narrow range 0.1399–0.1450, representing a

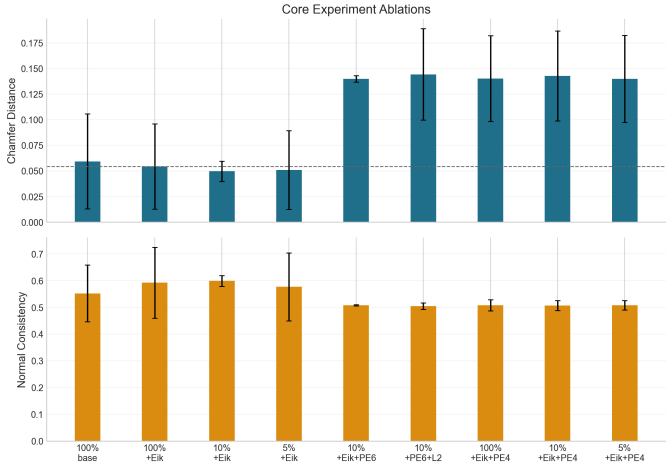

Fig. 3. Ablation across all experiments. Top: Chamfer Distance. Bottom: Normal Consistency. PE experiments (right group) cluster around CD ~0.14 regardless of configuration.

2.4–2.8× degradation from the non-PE results. This collapse is remarkably consistent: it occurs at every supervision level (5%, 10%, 100%), both PE frequencies ($L = 4$, $L = 6$), and even with additional second-order regularization (EXP-08).

Fig. 3 provides a comprehensive ablation view. The clear separation between PE and non-PE experiments confirms that positional encoding is the dominant factor degrading quality.

Normal Consistency further reveals the PE failure mode. Non-PE experiments show healthy NC variation across shapes (std ~0.10–0.14), while PE experiments exhibit collapsed NC diversity (std ~0.01–0.02). This indicates that PE forces all shapes toward similar, low-quality reconstructions.

C. PE Frequency Ablation

To test whether the PE failure was specific to $L = 6$, we conducted a follow-up sweep at $L = 4$ (EXP-10/11/12). As shown in Fig. 4, reducing PE frequency from $L = 6$ to $L = 4$ provides negligible improvement: CD changes by less than 3% across all supervision ratios, remaining firmly in the ~0.14 failure regime. The failure is not a matter of frequency tuning, suggesting a deeper incompatibility between Fourier PE and SDF evaluation via dense grid queries in this setup.

D. Per-Category Analysis

Table III breaks down the PE failure by shape category for two representative experiments. The category-level pattern is informative: airplanes consistently show the worst CD (~0.196), nearly double that of chairs (~0.124) and tables (~0.115). This ordering may reflect geometric complexity—airplanes have thinner structures and sharper features that are more sensitive to SDF extrapolation errors. Notably, Normal Consistency is nearly identical across all categories (~0.50–0.51), confirming that the PE collapse is universal rather than category-specific.

For non-PE experiments, per-category breakdowns were not recorded in the evaluation pipeline, which is a limitation of our

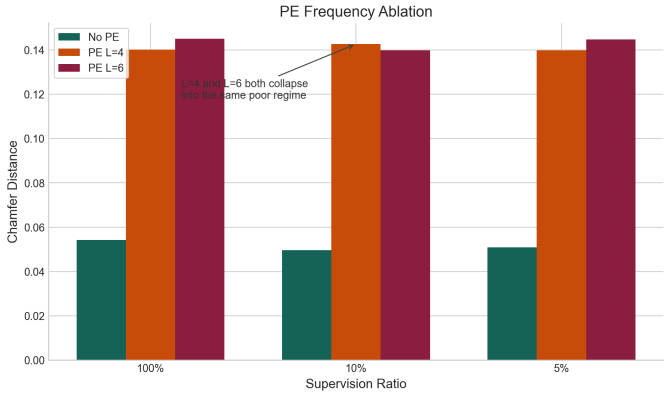

Fig. 4. PE frequency ablation. Both $L = 4$ and $L = 6$ collapse to the same poor Chamfer Distance regime at all supervision levels.

TABLE III
PER-CATEGORY CD AND NC FOR TWO PE EXPERIMENTS. THE PE FAILURE AFFECTS ALL CATEGORIES, WITH AIRPLANES SHOWING THE LARGEST DEGRADATION.

Experiment	Category	CD (↓)	NC (↑)
EXP-07 (5% + PE L=6)	Airplane	0.1957	0.5073
	Chair	0.1240	0.5081
	Table	0.1146	0.5066
EXP-09 (100% + PE L=6)	Airplane	0.1962	0.5023
	Chair	0.1242	0.5072
	Table	0.1147	0.4998

analysis. The per-shape CD standard deviations (0.02–0.05) in the non-PE experiments suggest meaningful category-level variation exists but was not captured.

E. Multi-Seed Validation

We validated our two key experiments with 3 seeds each (Table IV). EXP-04 (10% + Eikonal) shows moderate variance (CV = 0.197), primarily due to one seed (42) having higher CD with 37 mesh extraction failures; seeds 123 and 456 both achieve CD ~0.044 with zero failures. EXP-06 (10% + PE) shows very low variance (CV = 0.022), confirming the PE failure is highly reproducible.

The near-zero CV of EXP-06 is itself informative: PE is so dominant a failure mode that it overwhelms the natural stochasticity of training, producing nearly identical poor outcomes regardless of random initialization.

F. Note on Evaluation Pipeline

An initial version of our evaluator coupled metric computation with mesh decimation for export; when decimation failed (common for the large, noisy meshes produced by PE experiments), the per-shape metrics were discarded even though they had been computed successfully. This caused early PE runs to report 0/300 successful evaluations despite having valid CD and NC values. We fixed the evaluator to separate metric computation from mesh export and re-ran evaluation for all PE experiments (EXP-06 through EXP-12, which includes EXP-08). After the fix, valid CD and NC metrics were recovered for all 300 shapes in each PE experiment, even though the mesh-export success counts in the original logs had shown 0/300 due to the coupled decimation step. The non-PE single-seed experiments (EXP-01 through EXP-05) were not re-evaluated; of these, EXP-02, EXP-03, and EXP-05 had 39, 41, and 5 mesh extraction failures respectively, and their reported means are computed over the successfully evaluated subset only.

TABLE IV
MULTI-SEED VALIDATION FOR KEY EXPERIMENTS (3 SEEDS: 42, 123, 456). CV = COEFFICIENT OF VARIATION.

Experiment	CD mean	CD std	CV	NC mean
EXP-04 (10% + Eik)	0.0496	0.0098	0.197	0.5987
EXP-06 (10% + Eik + PE)	0.1399	0.0031	0.022	0.5078

VI. DISCUSSION

A. Why Eikonal Regularization Enables Label Reduction

The Eikonal constraint $\|\nabla_x f\| = 1$ provides a geometric prior that is complementary to direct SDF supervision. While $\mathcal{L}_{\text{sdf}}$ teaches the network the correct distance values at specific points, $\mathcal{L}_{\text{eik}}$ constrains the *gradient field* everywhere, enforcing that the learned function behaves as a valid distance function throughout the volume—not just at labeled locations.

Unsupervised points contribute to $\mathcal{L}_{\text{eik}}$ without requiring SDF labels, effectively providing "free" supervision about the geometric structure of the solution space. At low supervision ratios, the Eikonal loss dominates the training signal, acting as a strong regularizer that prevents overfitting to the sparse labeled points. One possible explanation for the counter-intuitive finding that 10% supervision outperforms 100% is that with abundant labels the network may overfit to noise in the SDF values, while with fewer labels the Eikonal loss contributes a larger share of the training signal, acting as a smoothness prior. However, this remains a hypothesis: the non-monotonic trend could also reflect interactions with the warmup schedule, batch composition, or other training dynamics. A definitive causal account would require controlled ablations isolating each factor.

The ratio-dependent warmup schedule is important: without it, the Eikonal loss can dominate too early before the network establishes basic geometry from $\mathcal{L}_{\text{sdf}}$, leading to degenerate solutions.

B. Why Fourier PE Fails for SDF Learning

The PE failure is surprising given its success in NeRF [3]. We hypothesize that it can be understood through the lens of train-test distribution mismatch. Training points are loaded from preprocessed samples whose spatial distribution is determined by the preprocessing pipeline; in typical SDF preprocessing, these points are concentrated near shape surfaces and within the unit sphere. Evaluation via marching cubes, however, queries the SDF on a dense 128^3 grid spanning

$[-1, 1]^3$, including cube corners at distance $\sqrt{3} \approx 1.73$ from the origin—potentially including regions underrepresented in the training distribution.

Without PE, the MLP's inductive bias toward smooth functions produces reasonable SDF extrapolation in these unseen regions. With PE, the sinusoidal encoding amplifies high-frequency oscillations, causing wild SDF fluctuations at grid points far from the training distribution. This creates spurious zero-crossings that produce noisy, oversized meshes during marching cubes extraction.

Several observations are consistent with this hypothesis:

- Training losses converge normally for PE experiments ($\mathcal{L}_{\text{sdf}} \sim 0.033$), indicating the failure is at inference, not training.
- NC standard deviation collapses to ~0.01 across all PE experiments, suggesting all shapes converge to similar poor-quality reconstructions.
- Reducing PE frequency from $L = 6$ to $L = 4$ does not resolve the issue, as even $L = 4$ introduces sufficient high-frequency components to corrupt extrapolation.
- Second-order regularization ($\mathcal{L}_{\text{2nd}}$) does not help because it penalizes curvature at training points, not at the distant grid points where PE extrapolation fails.

This finding highlights a key difference between radiance fields and distance fields: NeRF evaluates primarily along training camera rays, while SDF evaluation requires dense volumetric queries, making extrapolation robustness critical.

C. Limitations

Our study has several limitations. We evaluate on a single dataset (300 ShapeNet shapes across 3 categories), and results may differ with more diverse geometry or larger datasets. We use a fixed marching cubes resolution of 128^3; higher resolutions might reveal different PE behavior. IoU evaluation was omitted due to computational constraints. The non-monotonic label efficiency curve (5% slightly outperforming 10%) warrants further investigation with additional seeds and categories. Qualitative mesh comparisons are limited to non-PE experiments (Fig. 2); PE mesh exports were not retained (Section V-F). Finally, our PE analysis is limited to standard Fourier encoding; alternative approaches such as hash grid encodings [12] or learned feature grids may interact differently with SDF learning.

VII. Conclusion

We demonstrate that Eikonal regularization is an effective mechanism for reducing supervision requirements in DeepSDF. With only 10% of signed distance labels and Eikonal regularization on unsupervised points, we achieve a 3-seed mean Chamfer Distance that is competitive with, and in two of three seeds better than, the single-seed fully-supervised baseline across 300 ShapeNet shapes (see Section V-A). We note that our evaluation is on training shapes (not a held-out test set), so these results reflect fitting quality rather than generalization to novel shapes. Nonetheless, the consistent improvement across supervision ratios and the multi-seed validation suggest that competitive performance at 10% supervision is a promising finding with practical implications for scaling neural implicit reconstruction.

In contrast, Fourier positional encoding proves consistently detrimental in this setting, degrading Chamfer Distance by 2.4–2.8× regardless of supervision level, PE frequency, or additional regularization. We attribute this to PE's amplification of high-frequency artifacts in volumetric regions outside the training distribution, a failure mode specific to SDF evaluation via marching cubes.

For practitioners, the combination of Eikonal regularization with reduced supervision (10%) is a promising starting point for DeepSDF training, though our results show non-trivial seed variance at this ratio and further validation on held-out shapes and additional datasets would be needed before treating it as a reliable default. For future work, alternative input encodings that avoid Fourier features' extrapolation problems—such as multi-resolution hash grids [12] or learned spatial features—merit investigation. Additionally, extending the semi-supervised framework to category-specific evaluation and larger datasets would further validate the generality of these findings.

References

[1] J. J. Park, P. Florence, J. Straub, R. Newcombe, and S. Lovegrove, "DeepSDF: Learning continuous signed distance functions for shape representation," in *Proc. IEEE/CVF CVPR*, 2019, pp. 165–174.

[2] A. Gropp, L. Yariv, N. Haim, M. Atzmon, and Y. Lipman, "Implicit geometric regularization for learning shapes," in *Proc. ICML*, 2020, pp. 3789–3799.

[3] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," in *Proc. ECCV*, 2020, pp. 405–421.

[4] M. Tancik *et al.*, "Fourier features let networks learn high frequency functions in low dimensional domains," in *Proc. NeurIPS*, 2020, pp. 7537–7547.

[5] V. Sitzmann, J. N. P. Martel, A. W. Bergman, D. B. Lindell, and G. Wetzstein, "Implicit neural representations with periodic activation functions," in *Proc. NeurIPS*, 2020, pp. 7462–7473.

[6] A. X. Chang *et al.*, "ShapeNet: An information-rich 3D model repository," *arXiv preprint arXiv:1512.03012*, 2015.

[7] W. E. Lorensen and H. E. Cline, "Marching cubes: A high resolution 3D surface construction algorithm," *ACM SIGGRAPH Computer Graphics*, vol. 21, no. 4, pp. 163–169, 1987.

[8] L. Mescheder, M. Oechsle, M. Niemeyer, S. Nowozin, and A. Geiger, "Occupancy Networks: Learning 3D reconstruction in function space," in *Proc. IEEE/CVF CVPR*, 2019, pp. 4460–4470.

[9] L. Yariv *et al.*, "Volume rendering of neural implicit surfaces," in *Proc. NeurIPS*, 2021, pp. 4805–4815.

[10] P. Wang *et al.*, "NeuS: Learning neural implicit surfaces by volume rendering for multi-view reconstruction," in *Proc. NeurIPS*, 2021, pp. 27171–27183.

[11] N. Rahaman *et al.*, "On the spectral bias of neural networks," in *Proc. ICML*, 2019, pp. 5301–5310.

[12] T. Müller, A. Evans, C. Schied, and A. Keller, "Instant neural graphics primitives with a multiresolution hash encoding," *ACM Trans. Graphics*, vol. 41, no. 4, pp. 1–15, 2022.